

# CURRICULUM VITAE

## Contact information

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## Education

- PhD, Statistical Genetics (supervisor: Prof Pak Sham), King's College London, UK

## Professional positions

2005.9-2018.7 Investigator scientist in Genetics, MRC Epidemiology Unit

2018.8- Genetic Analyst / Senior Research Associate, Cardiovascular Epidemiology Unit, Department of Public Health and Primary Care, University of Cambridge

## Research

I work on human health-related research, over years evolving from familial aggregation, segregation, linkage, candidate genes and genomewide association studies (GWASs) to a recent focus on proteogenomics within the SCALLOP consortium using Olink and mass spectrometry (MS) data. I led collaborative analysis to the SCALLOP-Seq(ue), both WES and NGS) consortium and Host Genetics Initiative.

I promote reproducible research as in <https://jinghuazhao.github.io/software>, <https://jinghuazhao.github.io/Computational-Statistics/> and <https://jinghuazhao.github.io/Omics-analysis/>.

Integration of computational statistics, machine learning and artificial intelligence is shown from my creation of <https://cambridge-ceu.github.io/csd3/systems/ceuadmin.html>, involving aspects on

- AI: BitNet, Claude Code, Ollama, OpenClaw, codex-cli, copilot-cli, gemini-cli, hermes-agent, llama.cpp as with models, platforms and libraries such as TensorFlow, PyTorch, scikit-llm, LangChain;
- MS data: InstaNovo & DIA-NN
- Single-cell omics: Seurat, scp, scanpy, scvi-tools, scGPT, C2S-Scale;
- mtDNA analysis: MToolBox, fNUMT, haplogrep; long-read sequencing analysis: SVanalyzer, hap.py, sniffles, truvari;
- Population genetics: selscan, angsd, relate, clues2 and fastsimcoal2.

## Key publications

**Zhao JH**, et al. Mapping pQTLs of circulating inflammatory proteins identifies drivers of immune-mediated disease risk and novel therapeutic targets. *Nat Immunol* 2023, **24**(9):1540-1551, 10.1038/s41590-023-01588-w, <https://www.nature.com/articles/s41590-023-01588-w>.

COVID-19 Host Genetics Initiative. Mapping the human genetic architecture of COVID-19. *Nature* **600**:472–477 (2021); Pathak, G.A. et al. A first update on mapping the human genetic architecture of COVID-19. *Nature* **608**:E1-E10 (2022); The Host Genetics Initiative. A second update on mapping the human genetic architecture of COVID-19. *Nature* **621**:E7–E26 (2023).

**Zhao JH**, Luan JA, Congdon P. Bayesian linear mixed model of polygenic effects. *J Stat Soft.* 2018, **85**(6):1-27. doi: 10.18637/jss.v085.i06



**Zhao JH**, Luan JA. Mixed modeling with whole genome data. *J Prob Stat*. 2012. doi: 10.1155/2012.485174.

Xue F, Li S, Luan J, Yuan Z, Luben RN, Khaw K-T, Wareham NJ, Loos RJF, **Zhao JH**. A latent variable partial least squares path modeling approach to regional association and polygenic effect with applications to a human obesity study. *PLoS ONE* 2012, **7**(2): e31927

Loos RJ, *et al*. Common variants near *MC4R* are associated with fat mass, weight and risk of obesity. *Nat Genet* 2008; **40**(6):768-75

**Zhao JH**. gap: genetic analysis package. *J Stat Soft* 2007, **23** (8):1-18. doi: 10.18637/jss.v023.i08.

## Publications since 2018

188. Demenais F, *et al*. Multi-ancestry genome-wide association study identifies new asthma susceptibility loci that co-localize with immune cell enhancer histone marks. *Nat Genet* 2018; **50**(1):42-53.
189. Medina-Gomez C, *et al*. Life-course genome-wide association study meta-analysis of total body BMD and assessment of age-specific effects. *Am J Hum Genet* 2018; **102**(1):88-102.
190. Sung YJ, *et al*. A large-scale multi-ancestry genome-wide study accounting for smoking behavior identifies multiple significant loci for blood pressure. *Am J Hum Genet* 2018, **102**(3):375-400. DOI: 10.1016/j.ajhg.2018.01.015.
191. Turcot V, *et al*. (2018). Protein-altering variants associated with body mass index implicate pathways that control energy intake and expenditure underpinning obesity. *Nat Genet* **50**:26-41.
192. **Zhao JH**, Luan JA, Congdon P. Bayesian linear mixed model of polygenic effects. *J Stat Soft* 2018, **85**(6):1-27. DOI: 10.18637/jss.v085.i06.
193. Feitosa MF, *et al*. Novel genetic associations for blood pressure identified via gene-alcohol interaction in up to 570K individuals across multiple ancestries. *PLoS One* 2018, **13**(6):e0198166. DOI: 10.1371/journal.pone.0198166.
194. Lee JJ, *et al*. Gene discovery and polygenic prediction from a genome-wide association study of educational attainment in 1.1 million individuals. *Nat Genet* 2018, **50**:1112–1121, DOI: 10.1038/s41588-018-0147-3.
195. Ligthart S, *et al*. Genome analyses of >200,000 individuals identify 58 loci for chronic inflammation and highlight pathways that link inflammation and complex disorders. *Am J Hum Genet* 2018, **103**(5):691-706. DOI: 10.1016/j.ajhg.2018.09.009.
196. Evangelou E, *et al*. Genetic analysis of over 1 million people identifies 535 new loci for blood pressure traits. *Nat Genet* 2018, **50**(10):1412-1425. DOI: 10.1038/s41588-018-0205-x.
197. Merino J, *et al*. Genome-wide meta-analysis of macronutrient intake of 91,114 European ancestry participants from the cohorts for heart and aging research in genomic epidemiology consortium. *Mol Psychiatr* 2018 Jul 9. DOI: 10.1038/s41380-018-0079-4.
198. Kilpeläinen TO, *et al*. Multi-ancestry study of blood lipid levels identifies four loci interacting with physical activity. *Nat Comm* 2019, **10**, 376, DOI: 10.1038/s41467-018-08008-w.
199. Giri A, *et al*. Trans-ethnic association study of blood pressure determinants in over 750,000 individuals. *Nat Genet* 2019, **51**:51-62
200. Karasik D, *et al*. Disentangling the genetics of lean mass. *Am J Clin Nutr* 2019, **109**(2): 276-287, DOI: 10.1093/ajcn/nqy272.
201. de Vries PS, *et al*. Multi-ancestry genome-wide association study of lipid levels incorporating gene-alcohol interactions. *Am J Epidemiol* 2019, **188**(6):1033-1054, DOI: 10.1093/aje/kwz005.
202. Justice AE, *et al*. Protein-coding variants highlight the importance of lipolysis in adipocytes for body fat distribution. *Nat Genet* 2019, DOI: 10.1038/s41588-018-0334-2.

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203. Shrine N, et al. New genetic signals for lung function highlight pathways and chronic obstructive pulmonary disease associations across multiple ancestries, *Nat Genet* 2019, DOI: 10.1038/s41588-018-0321-7.
204. Zhao J, et al. Meta-analysis of genome-wide association studies provides insights into genetic control of tomato flavor. *Nat Comm* 2019, 10, Article number: 1534 (2019), DOI: 10.1038/s41467-019-09462-w.
205. Sung YJ, et al. A multi-ancestry genome-wide study incorporating gene–smoking interactions identifies multiple new loci for pulse pressure and mean arterial pressure. *Hum Mol Genet* 2019, doi: 10.1093/hmg/ddz070.
206. Clark DW, et al. Associations of autozygosity with a broad range of human phenotypes. *Nat Comm* 2019, NCOMMS-18-33232A-Z.
207. Bentley A.R., et al. Multi-ancestry genome-wide gene–smoking interaction study of 387,272 individuals identifies new loci associated with serum lipids. *Nat Genet* **51**, 636–648 (2019). DOI: 10.1038/s41588-019-0378-y.
208. Shah S, et al. Genome-wide association and Mendelian randomisation analysis provide insights into the pathogenesis of heart failure. *Nat Comm* 2020, 11(1):163. Published 2020 Jan 9. DOI: 10.1038/s41467-019-13690-5.
209. Surendran P, et al. Discovery of rare variants associated with blood pressure regulation through meta-analysis of 1.3 million individuals. *Nat Genet* 2020, 52(12):1314–1332, DOI: 10.1038/s41588-020-00713-x.
210. Lin W, Ji J, Zhu Y, Li M, Zhao J, Xue F, Yuan Z. PMINR: pointwise mutual information-based network regression – with application to studies of lung cancer and Alzheimer’s disease. *Front Genet* 2020, 11:556259, DOI: 10.3389/fgene.2020.556259.
211. Cuellar-Partide G, et al. Genome-wide association study identifies 48 common genetic variants associated with handedness. *Nat Hum Behaviour* 2021, 5: 59–70.
212. Gaziano L, et al. Actionable druggable genome-wide Mendelian randomization identifies repurposing opportunities for COVID-19. *Nat Med* 2021, 27:668–676.
213. Chen J, et al. The trans-ancestral genomic architecture of glycemic traits. *Nat Genet* 2021, 53:840–860.
214. COVID-19 Host Genetics Initiative. Mapping the human genetic architecture of COVID-19. *Nature* 2021.
215. Zhang Y, et al. Mendelian randomisation highlights hypothyroidism as a causal determinant of idiopathic pulmonary fibrosis. *EBiomed.* 2021, DOI: 10.1016/j.ebiom.2021.103669
216. Graham SE, et al. The power of genetically diverse individuals in genome-wide association studies of blood lipid levels. *Nature* 2021, **600**: 675–679, DOI:10.1038/s41586-021-04064-3.
217. Jin X, Zhang L, Ji J, Ju T, Zhao JH, Yuan Z. NeRiT -- Network regression in transcriptome-wide association studies. *BMC Genomics* (2022) 23:562, DOI: 10.1186/s12864-022-08809-w.
218. Pathak, G.A. et al. A first update on mapping the human genetic architecture of COVID-19. *Nature* **608**, E1-E10 (2022).
219. Ramdas S, et al. A multi-layer functional genomic analysis to understand noncoding genetic variation in lipids. *Am J Hum Genet.* 2022 Aug 4;**109**(8):1366–1387. DOI: 10.1016/j.ajhg.2022.06.012.
220. Wang Z, et al. Genome-wide association analyses of physical activity and sedentary behavior provide insights into underlying mechanisms and roles in disease prevention *Nat Genet.* 2022, **54**:1332–1344
221. Yengo L, et al. A saturated map of common genetic variants associated with human height from 5.4 million individuals of diverse ancestries. *Nature* 2022, **610**:704–712.
222. Kanoni, S., et al. Implicating genes, pleiotropy, and sexual dimorphism at blood lipid loci through multi-ancestry meta-analysis. *Genome Biology* 2022, **23**(1): 268 (medRxiv, DOI:10.1101/2021.12.15.21267852).

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224. van de Vegte JY, et al. Genetic insights into resting heart rate and its role in cardiovascular disease. *Nat Comm* **14**:4646 (2023), DOI: 10.1038/s41467-023-39521-2.
225. **Zhao JH**, et al. Mapping pQTLs of circulating inflammatory proteins identifies drivers of immune-mediated disease risk and novel therapeutic targets. *Nat Immunol* 2023, **24**(9):1540-1551, DOI: 10.1038/s41590-023-01588-w.
226. Lagou V, et al. GWAS of random glucose in 476,326 individuals provide insights into diabetes pathophysiology, complications and treatment stratification. *Nat Genet* **55**:1448–1461 (2023), DOI: 10.1038/s41588-023-01462-3, *medRxiv*, DOI: 10.1101/2021.04.17.21255471.
227. The Host Genetics Initiative. A second update on mapping the human genetic architecture of COVID-19. *Nature* **621**:E7–E26 (2023).
228. Macdonald-Dunlop E, et al. Mapping genetic determinants of 184 circulating proteins in 26,494 individuals to connect proteins and diseases. *medRxiv*, DOI:10.1101/2021.08.03.21261494.
229. Klaric L, et al. Mendelian randomisation identifies alternative splicing of the FAS death receptor as a mediator of severe COVID-19. *medRxiv*, DOI: 10.1101/2021.04.01.21254789.
230. Gaziano L, et al. Transcriptome- and proteome-wide Mendelian randomization to prioritize therapeutic targets for coronary heart disease. *medRxiv*, DOI: 10.1101/2024.06.27.24309406.
231. Kelemen M, et al. Performance of deep-learning based approaches to improve polygenic scores. *Nat Comm* **16**, 5122 (2025). DOI: 10.1038/s41467-025-60056-1.
232. Smit RAJ, et al. Polygenic scores to predict body mass index and obesity across populations and through the life course powered by data from 5.1 million individuals, *Nat Med* (2025). DOI: 10.1038/s41591-025-03827-z.
233. **Zhao JH** (2026, January 28). Genetic association analysis with R: latest developments. Published in The Biomedical & Life Sciences Collection, Henry Stewart Talks. DOI: 10.69645/FRFQ9519. ([HSTalks](https://hstalks.com)), <https://github.com/jinghuazhao/gaawr2>.
234. Koprulu M, et al. Multi-cohort proteogenomic analyses reveal genetic effects across the proteome and diseasome. *Cell* (2026). DOI: 10.1016/j.cell.2026.03.049.
235. Bo C, Li R, Zhao JH, Feng S, Ji X, Xue F. Agentic T-STROBE: A front-shifted STROBE paradigm for observational research.